

CoRoT-1b

R-2424

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_211120 · created 2026-08-22T00:00:16+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459538.9343 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.041
Transit depth (Rp/Rs)^2	2.0 +/- 1.16 [%]
Orbital Inclination (inc)	84.56 +/- 2.82
Airmass coefficient 1 (a1)	0.897 +/- 0.056
Airmass coefficient 2 (a2)	0.08 +/- 0.12
Scatter in the residuals of the lightcurve fit is	6.28 %
Best Comparison Star	#3 - [108, 113]
Optimal Aperture	2.92
Optimal Annulus	13.73
Transit Duration (day)	0.102 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.141 ± 0.041 vs 0.1388 (deviation 0.05σ)
Duration fit vs published	0.102 d vs 0.1046 d (deviation 0.18σ)
Mid-time O–C	11.8 min
Depth SNR	3.4
Residual scatter	6.28 %

Light curve

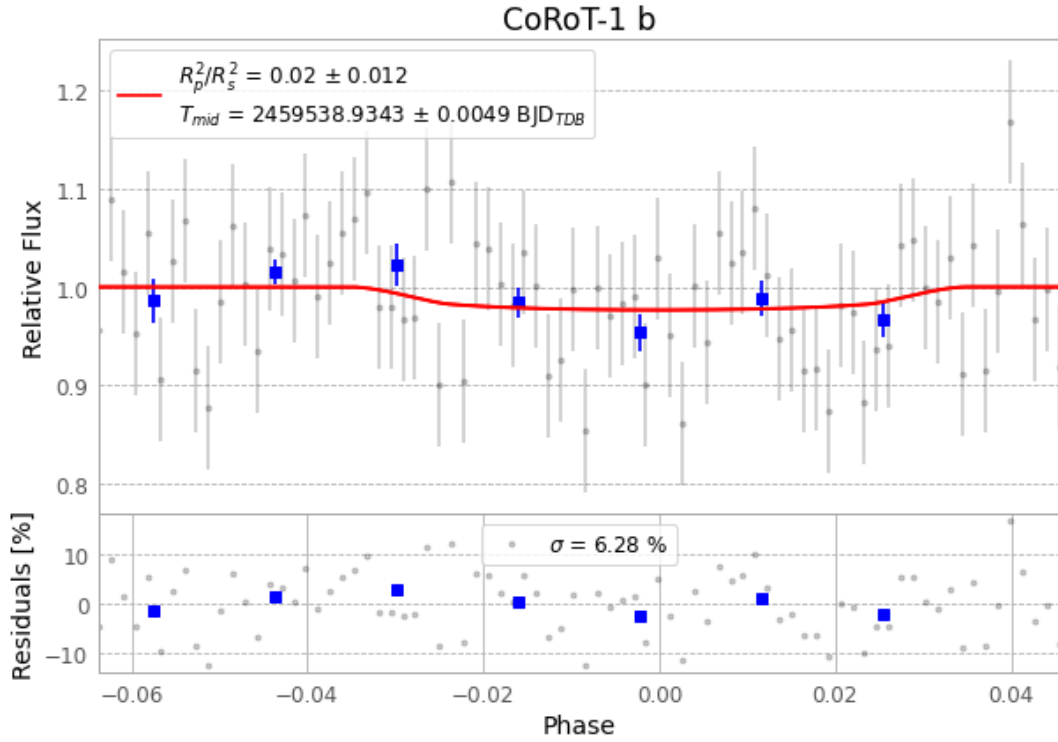


Figure 1 — FinalLightCurve_CoRoT-1 b_2021-11-20.png (SHA-256 459110e99d6eb8e4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [449, 156], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2a1213e92761ab3f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

80 FITS frames (53 MB), CoRoT-1211120080012.FITS → CoRoT-1211120115712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-211120.log (cc7e98041598...), FinalLightCurve_CoRoT-1 b_2021-11-20.png (459110e99d6e...), FinalLightCurve_CoRoT-1 b_2021-11-20.csv (2e9cdc284bac...)

Compute resources

Wall time 220.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 00:00Z.